

Infosys Springboard Virtual Internship 7.0 Completion Report

Project: Agentic AI for Maritime Freight Pricing and Route Optimization

Team Details

Batch Number: 2

Project Start Date: 21 July 2026

Internship Duration: 8 Weeks

Team Members: T. Rohitha; Aparajita De; Divyajyot Diwakar Sinha; Mohamad Nasthaeen; Shaik Baheed

1. Project Title

Agentic AI for Maritime Freight Pricing and Route Optimization

2. Project Objective

The project aims to build a smart maritime freight platform that brings shipment enquiry, route planning, freight pricing, quotation generation, operational tracking and role-based workflows into one system. In Milestone 3, the platform was extended with an agentic intelligence layer covering weather assessment, customs and regulatory intelligence, shipment-risk scoring and machine-learning pricing evaluation. The overall goal is to make freight decisions faster, more explainable and more risk-aware while keeping human review in the loop for compliance-sensitive cases.

Milestone 1 established the shipment and route foundation, Milestone 2 converted that foundation into a structured commercial pricing and quotation workflow, and Milestone 3 connected the workflow to intelligence services and decision gates. The Milestone 3 work completed during 23–28 August 2026 is documented separately below so that the report clearly distinguishes implemented capabilities from earlier roadmap concepts.

3. Project Description in Detail

Freight operations often involve multiple inputs such as origin and destination ports, transport mode, cargo details, container type, weight, Incoterms and commercial requirements. The project converts these inputs into a guided digital workflow so that a user can prepare a shipment request,

review route information and generate a structured quotation instead of relying only on manual calculations.

The current implementation is a React and Vite application written in TypeScript. It includes an Express-based backend layer with REST-style endpoints for pricing, quotes, master data, feedback and health checks. The interface provides separate role-oriented experiences for users, business/commercial teams, freight agents, brokers and system administrators. The application also supports PDF quotation generation and a quotation-history workflow.

Milestone 1 established the route and shipment foundation. Milestone 2 expanded this into a commercial pricing workflow with rate cards, surcharges, cost breakdown, margin policy, approvals, quote versions and broker review. Milestone 3 added the agentic intelligence layer: weather intelligence, customs/RAG checks, a weighted five-pillar risk engine, ML-versus-rule pricing evaluation, human-in-the-loop customs sign-off and an end-to-end orchestration flow. These additions build directly on the earlier route and pricing modules rather than replacing them.

Technology Stack

Layer / Area	Technology / Component	Purpose
Frontend	React 19 + TypeScript + Vite	Builds the responsive user, broker, agent and admin interfaces.
Styling	Tailwind CSS	Responsive layouts, dashboards, cards, forms and role-specific UI styling.
Backend	Node.js + Express + TypeScript	Provides REST-style application and pricing/quotation endpoints.
Pricing Engine	TypeScript pricing modules	Rate-card resolution, surcharges, cost build-up and margin enforcement.
Client Persistence	Browser localStorage	Stores seeded/registered users, drafts and selected client-side workflow state.
PDF	jsPDF	Generates printable/downloadable quotation documents.
Maps / Tracking UI	Google Maps React integration	Supports route and tracking-oriented visualizations in the application.
Deployment	Vercel	Hosts the deployed web application and serverless-compatible API entry points.
ML / Data Science	Python + NumPy + Pandas + LightGBM / XGBoost / Scikit-Learn	Training, feature engineering and evaluation of freight-price regression models.
Intelligence Layer	Weather Agent + Customs/RAG Agent + 5-Pillar Risk Engine	Produces weather, regulatory and explainable risk signals for freight decisions.

4. Timeline Overview

The supplied schedule information identifies the overall project start as 21 July 2026. Milestone 1 ran from 27 July to 7 August, Milestone 2 ran from 10 August to 21 August, and Milestone 3 was carried out from 23 August to 28 August 2026. The report retains these dates as supplied by the team.

Week	Activities Planned	Activities Completed / Status
Week 1	Project kickoff, requirements understanding, application structure and shipment workflow setup.	Project initiated on 21 July; core application structure and role-oriented flow established.
Week 2	Milestone 1 route and shipment foundation.	Shipment input, route operations, transport-mode selection and route-oriented UI implemented.
Week 3	Milestone 2 pricing foundation and commercial calculation flow.	Rate-card and pricing modules, cost build-up and quotation calculation workflow developed.
Week 4	Milestone 2 quote generation, margin/approval flow and broker review.	Quote versions, margin checks, approval endpoints, broker review and quotation history workflow completed.
Week 5	Milestone 3 intelligence integration and agent workflow.	Weather, customs/RAG, risk scoring, ML evaluation and role-separated compliance workflow implemented/validated. 23–28 August 2026.
Week 6	Milestone 3 hardening, testing and end-to-end verification.	80/80 documented test scenarios passed according to the supplied M3 test plan; defensive error handling and workflow integration verified.
Week 7		
Week 8		

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project team began the internship project and aligned on the maritime freight pricing and route optimization objective.	21 July 2026
Milestone 1 — Route & Shipment Foundation	Core shipment intake, transport-mode selection, route operations and route-oriented workflow established.	7 August 2026
Milestone 3 — Agentic Intelligence & Risk-Aware Pricing	Weather intelligence, customs/RAG validation, five-pillar shipment risk scoring, ML-vs-rule pricing evaluation, role-separated customs sign-off and end-to-end agentic workflow integrated and tested.	28 August 2026

Milestone 2 — Pricing & Quote Generation	Commercial pricing engine, cost build-up, margin policy, quotation workflow and broker review flow established.	21 August 2026
Final Submission	Final date was not provided in the supplied project information.	To be confirmed
Presentation	Presentation date was not provided in the supplied project information.	To be confirmed

5b. Project Execution Details

Milestone 1 — 27 July to 7 August 2026

Milestone 1 focused on establishing the operational foundation of the application. The user flow begins with role selection and authentication, then moves into shipment enquiry and route-oriented operations. The project source contains a dedicated Milestone1RouteOperationsView that supports shipment search/filtering, route status, route options, carrier information, transit information and selection of a route option before moving the shipment towards pricing.

The shipment form captures important freight parameters such as origin and destination port codes, transport mode, ocean load type, Incoterm, cargo items, container specification, quantity, gross weight, commodity description and HS code. This creates the structured input required for later pricing and intelligence layers.

The milestone also established role-aware navigation. The application distinguishes user, business, freight-agent, broker and admin experiences, allowing the same platform to support different operational responsibilities without presenting every tool to every role.

Milestone 2 — 10 August to 21 August 2026

Milestone 2 converted the shipment workflow into a commercial quotation workflow. The pricing layer includes rate-card resolution, air-weight-break logic, surcharge rules, Incoterm adjustments, special-handling charges, insurance, discounts, cost build-up and margin-policy enforcement.

The Milestone2QuoteBuilder calls the pricing API for a detailed cost breakdown and falls back to the local pricing engine when the API is unavailable. The quote result contains cost components, total cost, selling price, applied margin and approval requirements. If the requested margin falls below the configured policy floor, the quote is marked for approval rather than silently accepting the lower margin.

The quotation workflow also supports versioning, approval submission, approval/rejection, issuing a quote with a 14-day validity period, customer acceptance/decline, a customer-safe quote projection and feedback collection. The broker review interface can manually adjust base freight, BAF, terminal handling, documentation, broker margin and assigned carrier before final dispatch.

Milestone 3 — 23 August to 28 August 2026

Milestone 3 focused on turning the earlier freight quotation workflow into an intelligence-assisted decision flow. The project documentation defines a service-oriented architecture in which the customer/shipper portal, customs officer workspace, administrator console and shared risk/pricing views communicate with an Express API layer. The API layer exposes weather, customs, risk, pricing

and quotation services so that the different intelligence components can work together without changing the existing commercial quotation workflow.

The main objective was to add specialized agents and decision engines around the existing shipment and pricing flow. Instead of treating every shipment in the same way, the system can now evaluate route weather, customs requirements, shipment risk and the difference between deterministic pricing and ML-based price prediction. The design also keeps a human-in-the-loop path for compliance-sensitive shipments.

Agentic Workflow Implemented

The Milestone 3 workflow follows the sequence: shipment creation → route selection → pricing → weather assessment → customs validation → risk evaluation → quote preview → compliance queue when required → quotation/certificate generation. Weather and customs analysis can be triggered in parallel, after which the five-pillar risk engine combines their signals with route, port-congestion and cargo factors. The final decision determines whether a quote can be released automatically or needs officer review.

Weather Intelligence Agent

The Weather Intelligence Agent assesses maritime conditions along a route using waypoint-based route analysis. The documented design considers wind speed, swell/wave height, storm risk and seasonal effects, then converts the result into a delay probability and buffer recommendation. The system also supports an alternative-corridor advisory when conditions are severe. The design documents identify NOAA WaveWatch III, IMD maritime bulletins and Copernicus Marine Service as primary telemetry sources, with seeded corridor data available as an offline fallback.

Customs & Regulatory Intelligence Agent

The Customs Agent validates six- to eight-digit HS classifications, evaluates statutory duties such as BCD, IGST and SWS, checks restricted/prohibited cargo conditions and prepares a mandatory-document checklist. The project uses a Hybrid RAG design: lexical BM25 matching is combined with dense-vector similarity so that exact HS-code terms and broader regulatory meaning can both be retrieved. The documented knowledge sources include DGFT Foreign Trade Policy material, the CBIC Customs Tariff Act, the IMO IMDG Code and ICEGATE filing requirements. Recommendations are designed to retain source citations for traceability.

Five-Pillar Shipment Risk Engine

Milestone 3 introduced an explainable composite risk score from five factors: Weather 30%, Customs & Compliance 25%, Route & Corridor 20%, Port Congestion 15% and Cargo Fragility/Hazmat 10%. The resulting score is mapped to LOW (0–30), MEDIUM (31–60), HIGH (61–80) or CRITICAL (81–100). High-risk or restricted shipments can be routed to a Customs Officer workflow instead of being released automatically.

Machine-Learning Pricing Evaluation

The ML pricing work was documented as a supervised regression pipeline using a 5,000-observation freight-pricing dataset. Pandas is used for ingestion and cleansing, while NumPy/Pandas feature

engineering derives ton-kilometres, fuel-burn index, chargeable weight, stowage density, corridor velocity and logarithmic distance/weight features. The documented training process uses an 80/20 train-test split. The model-evaluation document reports LightGBM as the strongest benchmark in the comparison, with R^2 0.9745, MAE ₹2,420.80, RMSE ₹3,880.60, MAPE 3.42% and five-fold cross-validation R^2 0.976 ± 0.003 ; XGBoost and Scikit-Learn Gradient Boosting are also recorded as comparison models.

Human-in-the-Loop Customs Review

A key Milestone 3 design decision is that shippers do not approve their own regulatory checks. Shippers submit compliance information and can view status, while a dedicated Customs Officer workspace handles approval, document requests or rejection. This separation provides a realistic compliance workflow and preserves an audit-friendly human decision point for non-trivial cases.

Testing and Verification

The supplied Milestone 3 test plan records unit testing for the risk formula, duty calculation and weather interpolation; integration testing for the multi-agent workflow; role-isolation testing for customs sign-off; and defensive error handling for empty or null data. The documented test summary reports 12/12 Weather Agent tests, 16/16 Customs/RAG tests, 14/14 Risk Engine tests, 8/8 ML benchmark tests, 10/10 role/sign-off tests and 20/20 defensive null-safety tests passed, for an overall documented result of 80/80 scenarios validated.

Area	Implementation / Result	M3 relevance
Weather Intelligence	Route-waypoint assessment, storm-risk calculation, delay probability and buffer/advisory output.	Core agent
Customs + Hybrid RAG	HS validation, duty/tax calculation, document checklist, regulatory retrieval and citations.	Core agent
Risk Engine	Weighted five-pillar composite score with LOW/MEDIUM/HIGH/CRITICAL classification.	Decision gate
ML Pricing	Rule-versus-ML comparison with regression metrics and feature engineering pipeline.	Pricing intelligence
Role Separation	Dedicated Customs Officer review/sign-off path separate from shipper risk view.	Human-in-the-loop
End-to-End Flow	Shipment → route → pricing → weather → customs → risk → quote/compliance decision.	Integrated workflow

Testing	80/80 scenarios reported passed in the supplied M3 test plan.	Verification
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Core Pricing Flow

Step	What the system does
1. Shipment inputs	Reads route, mode, container, weight and Incoterm details.
2. Rate resolution	Looks for customer contract, general contract, carrier spot, published tariff, then predicted fallback.
3. Weight / charge basis	Uses container, CBM, chargeable weight or tonnage depending on transport mode.
4. Surcharges	Applies applicable BAF, CAF, terminal, security, documentation, customs and haulage rules.
5. Special handling	Adds charges for fragile, hazardous or temperature-controlled cargo where applicable.
6. Margin policy	Calculates the requested margin and checks it against the policy floor.
7. Approval	Routes below-floor or predicted-component cases into an approval workflow.
8. Quote issuance	Creates a quote version and, when issued, starts the 14-day validity clock.
9. Broker review	Allows commercial review, manual rate adjustment, carrier assignment and finalization.
10. Customer delivery	Presents a customer-safe quote, PDF output and acceptance/decline options.

6. Snapshots / Screenshots

The following screenshots were supplied with the project material and are included as visual evidence of the implemented application workflow. Captions distinguish completed M1/M2 screens from later-stage intelligence concepts.

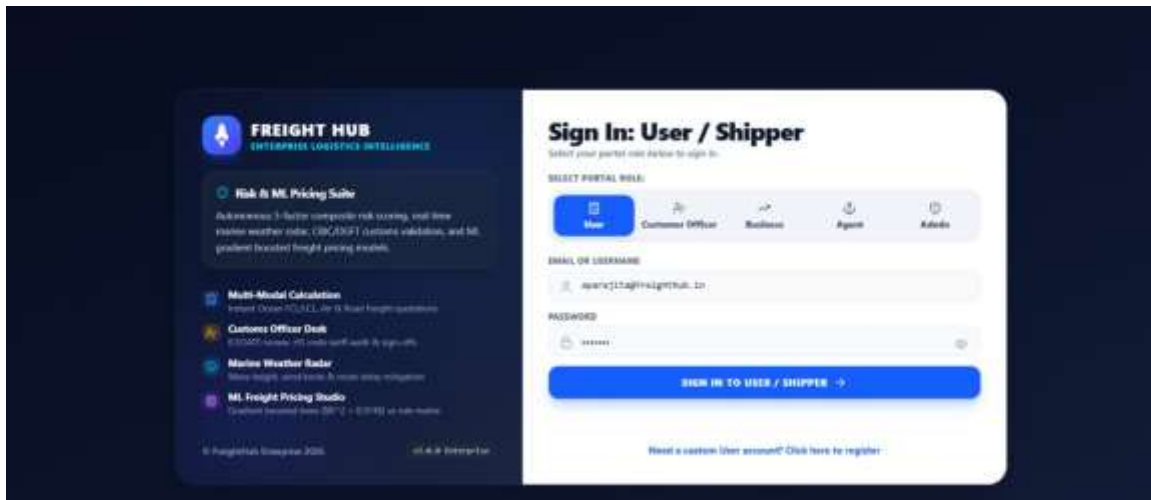


Figure 1: Role-based sign-in page for the User, Customer Officer, Business, Agent and Admin portals.

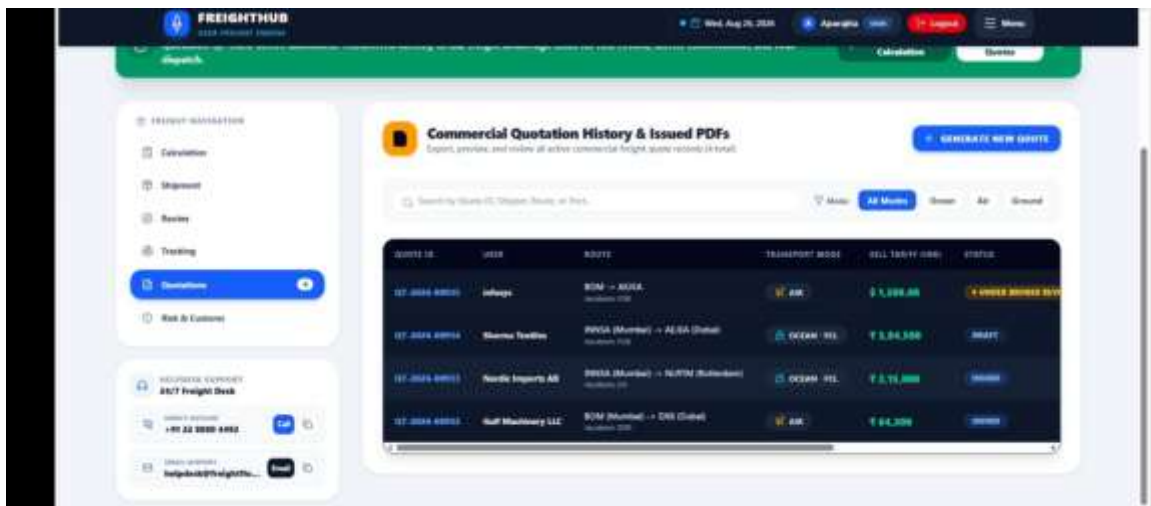


Figure 2: Commercial Quotation History & Issued PDFs page showing quote IDs, users, routes, transport modes, sell tariffs and quote statuses.

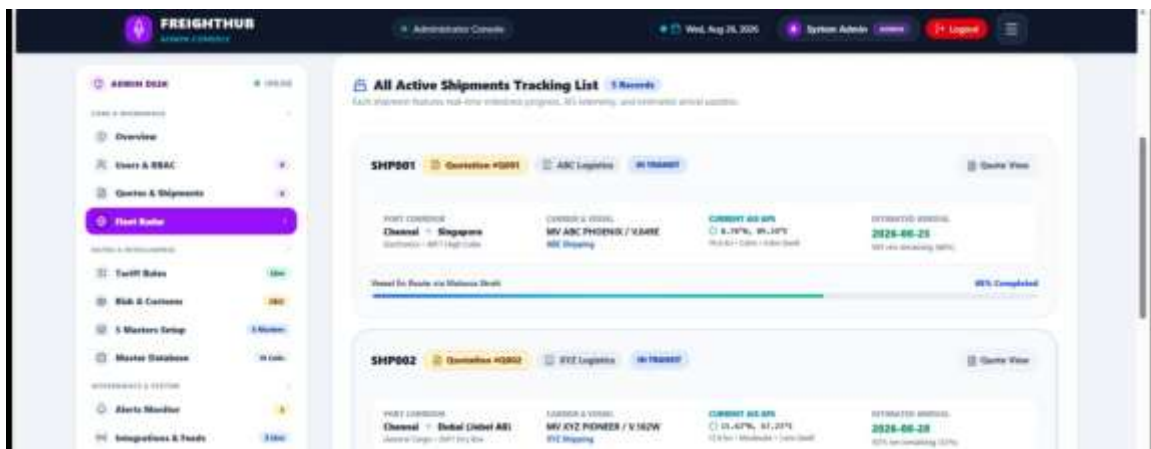


Figure 3: System Admin console showing the All Active Shipments Tracking List, including port corridors, carrier and vessel details, AIS GPS data, estimated arrival and transit progress.

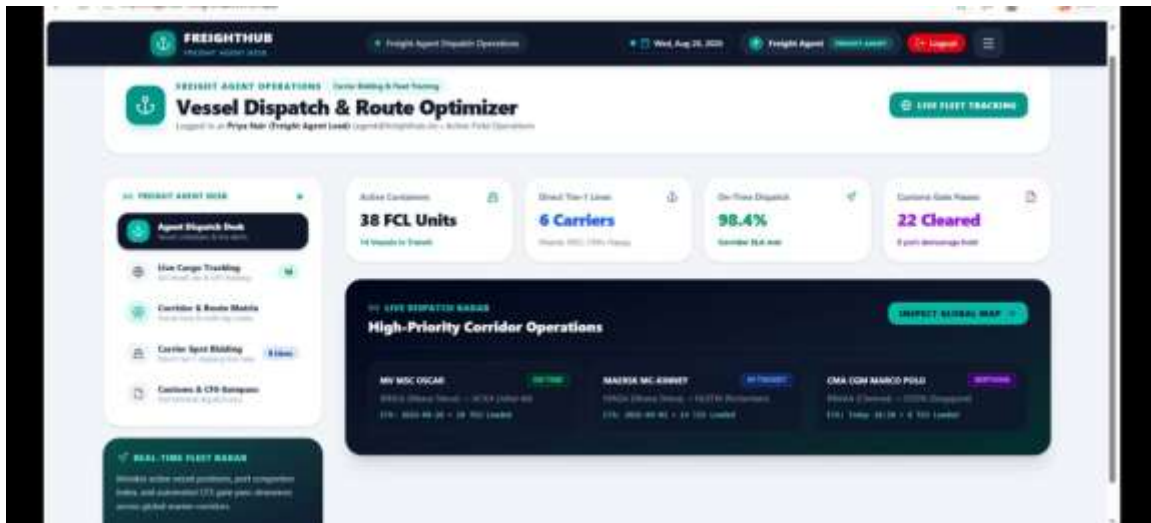


Figure 4: Freight Agent Desk – Vessel Dispatch & Route Optimizer showing active containers, Tier-1 carrier lines, on-time dispatch, customs gate passes and live high-priority corridor operations.



Figure 5: Official Freight Quotation summary (Estimate Ready) showing the \$1,208.88 all-inclusive tariff for the Mumbai (BOM) to Dubai (AEJEA) shipment, with AIR FCL service and a 14-day rate lock.

FreightHub
SMART LOGISTICS ENGINE - COMMERCIAL FREIGHT QUOTATION

QUOTE NO: QT-2026-00935
DATE ISSUED: 2026-08-26
STATUS: PENDING SHIPPER REVIEW

SUPPLIER / COMMERCIAL CLIENT
Infosys
Contact: Chyaguet
Email: abc@gmail.com
Country: India

ROUTE & SERVICE REQUIREMENTS
BOM -> AEJEA
Mode: AIR (FCL)
Incoterms: FOB
Est. Transit: 2-4 d (3,500 km (Airway))

CARGO PARAMETERS & SPECIFICATIONS

#	COMMODITY	PACKING / SPEC	QTY	GROSS WEIGHT	HS CODE
01	cotton	20GP	2	500 kg	920

Buttons: Back to Form, COMMERCIAL QUOTATION, QT-2026-00935, Print, Download PDF

Buttons: Back to Calculation Form, Print Quote, Download Official PDF

Figure 6: Issued Commercial Quotation details for Quote QT-2026-00935, showing the Infosys shipper, BOM → AEJEA route, AIR FCL service, FOB Incoterm, cargo specifications and confirmation status.

ITEMIZED COMMERCIAL TARIFF BREAKDOWN

Base AIR Tariff	\$ 888.00
Surinder Adjustment Factor (SAF) / Fuel Surcharge	\$ 84.00
Terminal Handling Charges (THC)	\$ 152.00
Port Documentation Filing & Customs Paperwork	\$ 96.00
Special Goods Handling Surcharge	\$ 35.00
Promotional Offer Discount	-\$ 126.00
TOTAL COMMERCIAL TARIFF:	\$ 1,208.88

COMMERCIAL SLA & TERMS:

- Quotation valid for 14 calendar days from date of issuance.
- Final rates subject to carrier, weather, delays and port customs clearance verification.
- Rates include import customs duties and destination demurrage charges if incurred.

Buttons: Back to Calculation Form, Print Quote, Download Official PDF

Figure 7: Itemized Commercial Tariff Breakdown showing the base air tariff, BAF/fuel surcharge, THC, documentation and customs paperwork, special handling surcharge, promotional discount and the resulting \$1,208.88 total.

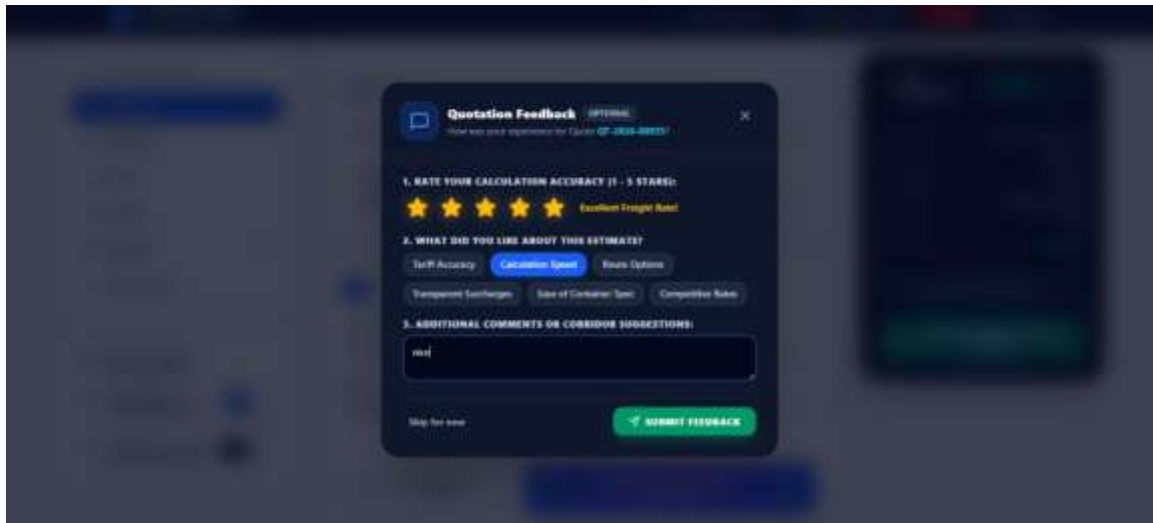


Figure 8: Quotation Feedback prompt showing the 1–5 star calculation-accuracy rating, feedback options, optional comments and submission control.



Figure 9: Swarm Calibration workflow for calculating freight tariff and routes, showing staged Pre-Flight Validation, Route Intelligence, Pricing Engine, Weather, Customs and Risk components.

- React component design, state management and role-based UI composition.
- TypeScript interfaces and modular separation of frontend, backend, pricing and data concerns.
 - REST-style API design using Express and JSON request/response flows.
- Freight-domain concepts such as Incoterms, FCL/LCL, chargeable weight, rate cards, BAF, THC, documentation charges and commercial margin.
 - Designing deterministic business rules for pricing, approvals and quote validity.
- Creating operational dashboards for administrators, brokers, freight agents and compliance-oriented roles.
- Understanding ML regression evaluation, feature engineering with NumPy/Pandas, and comparison of rule-based and predictive freight pricing.

- Understanding agent orchestration, Hybrid RAG, explainable risk scoring, human-in-the-loop compliance and end-to-end testing.

7. Challenges Faced

Managing a multi-role application: The project contains different portals for users, business/commercial teams, freight agents, brokers, customs officers and administrators. Milestone 3 added another layer of role separation by ensuring that customs sign-off actions are restricted to the dedicated officer workflow.

Keeping pricing and intelligence decisions consistent: Freight pricing combines deterministic business rules with predictive estimates, while risk combines several independent signals. The solution separates the rule engine, ML benchmark, weather, customs and risk components so each can be tested and explained independently.

Handling compliance-sensitive decisions: Not every shipment should be automatically cleared. Milestone 3 introduces explicit risk levels and a human-in-the-loop customs path so that high-risk or restricted cases can be reviewed before quotation release.

Balancing live intelligence with reliable fallbacks: Weather and regulatory services may depend on external data. The design therefore documents seeded/offline corridor data and structured service responses so the workflow can still be demonstrated and tested consistently.

Maintaining explainability: The project does not only return a final price or risk label. It exposes itemized pricing, weighted risk factors, regulatory citations and ML-versus-rule comparisons so that users and reviewers can understand why a decision was made.

Keeping milestone scope clear: The repository contains both implemented M1/M2 commercial workflows and M3 intelligence components. The final report therefore records the M3 functionality from the supplied M3 documents and does not present unrelated future ideas as completed deliverables.

8. Learnings & Skills Acquired

- React component design, state management and role-based UI composition.
- TypeScript interfaces and modular separation of frontend, backend, pricing and data concerns.
- REST-style API design using Express and JSON request/response flows.
- Freight-domain concepts such as Incoterms, FCL/LCL, chargeable weight, rate cards, BAF, THC, documentation charges and commercial margin.
- Designing deterministic business rules for pricing, approvals and quote validity.
- Creating operational dashboards for administrators, brokers and freight agents.
- Generating quotation PDFs and presenting itemized cost breakdowns clearly.
- Working collaboratively on a multi-module internship project and converting milestone requirements into demonstrable UI workflows.

9. Testimonials from Team

The team experience centered on converting a complex logistics problem into a clear digital workflow and then extending that workflow with intelligence. The progression from shipment and

route handling in Milestone 1, to commercial quotation in Milestone 2, and finally to weather, customs, risk and ML-assisted decision support in Milestone 3 made the project easier to understand and demonstrate.

Milestone 3 also helped the team understand that an AI-enabled freight platform must balance automation with accountability. Pricing should remain commercially explainable, regulatory decisions should retain human oversight where needed, and risk signals should be visible rather than hidden behind a single score.

10. Conclusion

Across the three milestones, the project progressed from a shipment and route foundation to a commercial quotation workflow and then to an integrated intelligence layer. Milestone 1 established structured shipment inputs, route operations and role-aware navigation. Milestone 2 added rate cards, surcharges, margin governance, approval flow, quote versioning, broker review and quotation documents. Milestone 3 added weather intelligence, customs/RAG validation, five-pillar risk scoring, ML pricing evaluation, human-in-the-loop customs review and an end-to-end agentic workflow.

The resulting system provides a strong prototype for agent-assisted maritime freight pricing and route optimization. The project architecture keeps commercial rules, predictive models, regulatory knowledge and operational dashboards modular, making the platform easier to test and extend. The supplied Milestone 3 documents and test plan provide the evidence basis for the completed intelligence work described in this report.

11. Acknowledgements

The team sincerely thanks Infosys Springboard for providing the internship environment and project opportunity. We also acknowledge the support and guidance received during the project, as well as the collaborative effort of all five team members: Aparajita De, T. Rohitha, Divyajyot Diwakar Sinha, Mohamad Nasthaeen and Shaik Baheed.

Appendix A — Key Application Modules

Module	Role / Purpose	Milestone relevance
Authentication & Role Selection	Provides role-aware sign-in and routes users to the appropriate workspace.	M1 foundation
Calculation Form	Collects origin, destination, mode, cargo, container, weight, HS code and Incoterm inputs.	M1 → M2
Route Operations	Searches shipment records, shows route status, carrier and transit details, and supports route selection.	M1
Pricing Engine	Builds itemized freight cost using rate cards, surcharges, Incoterms and other rules.	M2
Quote Builder	Creates quote versions, calculates selling price and records approval requirements.	M2
Broker Review	Allows manual rate adjustment, margin selection, carrier assignment and final dispatch.	M2

Quotation History / PDF	Stores and displays quote records and generates quotation documents.	M2
Admin Console	Provides operational oversight, user management, tariff/master-data tools and shipment tracking views.	M1/M2 support
Freight Agent Desk	Supports vessel dispatch, fleet tracking, carrier information and route operations.	Operational support
Agentic AI Preview	Shows the intended orchestration of pricing, route, weather, customs and risk agents.	Future roadmap / later milestone
Weather Intelligence Agent	Assesses route weather, storm risk, delay probability and buffer/advisory information.	M3
Customs & RAG Agent	Validates HS codes, duties, restrictions and mandatory documents using Hybrid RAG with citations.	M3
5-Pillar Risk Engine	Combines weather, customs, route, port and cargo factors into an explainable composite score.	M3
ML Pricing Benchmark	Compares rule-based pricing with regression models and records evaluation metrics.	M3
Customs Officer Workspace	Provides human review, document verification and sign-off for compliance-sensitive cases.	M3

Appendix B — Main API Endpoints Evidenced in the Source

Endpoint	Method	Purpose
/api/health	GET	Backend health check.
/api/v1/pricing/rate-cards	GET	Returns active rate-card records.
/api/v1/pricing/rate-cards/validate	POST	Validates an imported rate-card payload before commit.
/api/v1/pricing/rate-cards/commit	POST	Commits validated rate-card lines.
/api/v1/pricing/cost-breakdown	GET	Builds the detailed commercial cost breakdown.
/api/v1/quotes	POST	Creates a quote version.
/api/v1/quotes/:id/versions	GET	Retrieves quote versions.
/api/v1/quotes/:id/margin	PATCH	Updates margin subject to the margin-floor rule.
/api/v1/quotes/:id/submit-approval	POST	Submits a quote for managerial approval.
/api/v1/quotes/:id/approve	POST	Approves a quote.

/api/v1/quotes/:id/reject	POST	Rejects a quote with a reason.
/api/v1/quotes/:id/issue	POST	Issues a quote and starts its 14-day validity clock.
/api/v1/quotes/:id/accept	POST	Records customer acceptance.
/api/v1/quotes/:id/decline	POST	Records customer decline.
/api/feedback	GET / POST	Retrieves or records quotation feedback.
/api/v1/master-data/...	GET / POST / PUT / DELETE	Manages master-data collections used by the application.

Appendix C — Source Evidence and Scope Note

Primary implementation evidence used for this report: the supplied agentic_freighthub-main.zip repository, its docs folder containing the Milestone 3 requirements, architecture, dataset/model, API, testing and design documents, the supplied project screenshots, the supplied internship completion-report template, and the deployed application at <https://agentic-freighthub.vercel.app/>.

The repository contains the implemented M1/M2 commercial workflows and the documented M3 intelligence components. The Milestone 3 documentation specifically records the weather agent, customs/RAG agent, five-pillar risk engine, ML pricing evaluation, role-separated customs workflow, API contracts, data schemas and an end-to-end test plan. Where the repository contains design or seeded-data material rather than evidence of an external production service, this report describes it as documented/validated project functionality rather than as a claim of live third-party production deployment.